

Section A: Defining Medication Available Period (MAP) medications

Time Period	Variable	Label	Values	Definition	Code Lists Group	Additional Notes	Date Modified (Initials + Date)	QC Reviewed (Initials + Date)
FU_PD	LOT1_END_DT_TEMP	Temporary end date for LOT1 base period		The LOT will continue until the earliest of any of the following: (1) Discontinuation of all agents in the regimen (with or without switch to a new agent) - the LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available; (2) SCT / CAR-T events: if an autologous SCT occurs > 180 days following a previous autologous SCT, the LOT will end the day before the latter SCT. Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before the allogeneic SCT. CAR-T cellular therapy infusions are classified as their own LOT; the preceding LOT transition for CAR-T is defined in the LOT1_TX_ENDDATE_REASON and LOT1_BASE_END_DT rows. (3) Death; (4) Health plan disenrollment; (5) End of the study period.		LOT1_BASE_END_DT derived from LOT1_BASE_DISCON_DT (max MAP_END_DT from T_MEDICAL_PROG_CD/NDC and T_RX_NDC for induction meds), LOT1_BASE_1ST_ADD_MED_DT (first MAP_START_DT from T_MEDICAL_TX_RX for non-induction meds minus 1), LOT1_TX_ENDDATE (earliest LOT-ending SCT from T_MEDICAL procedure codes minus 1), and OBS_END_DT (LEAST of study end and config.DEATH_DT from T_DOD_YNDOD_disenrollment from T_MEMBER_CONF_T_ENROLLMENT_ELIGEND).		JM 15Apr 2026
				Note: Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT (this is an exception, not an ending event). The LOT will continue until the earliest of any of the following: (1) Discontinuation of all agents in the regimen, with or without switch to a new agent. The LOT end date is the run-out date, defined as the last date that any MM medication in that LOT is considered to be available. If the current LOT is immediately interrupted by a new agent, the LOT end date is the day before the first administration / dispense date of the new agent. (2) SCT / CAR-T events: if an autologous SCT occurs > 180 days following a previous autologous SCT, the LOT will end the day before the latter SCT. Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before the allogeneic SCT. CAR-T cellular therapy infusions are classified as their own LOT; the preceding LOT transition for CAR-T is defined in the LOT1_TX_ENDDATE_REASON and LOT1_BASE_END_DT rows. (3) Death; (4) Health plan disenrollment; (5) End of the study period.		End reason derived from matching the LOT1 end date against dates from: T_MEDICAL_TX_RX (discontinuation via MAP_END_DT), T_MEDICAL_TX (new med via MAP_START_DT), T_DOD (death via YNDOD), T_MEDICAL (SCT via procedure codes in T_MED_PROCECDURE), ENDDATE from T_MEMBER_CONF_T_ENROLLMENT (disenrollment) and study_end config.		JM 06Apr 2026
FU_PD	LOT1_END_REASON_TEMP	Temporary LOT1 base period end reason						
FU_PD	LOT1_HSCT_AUTO_SING_FL	Flag: base period had single valid autologous HSCT		Note: Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT (this is an exception, not an ending event). Stem cell transplant (SCT): A LOT-specific binary variable is created to classify whether patients have an SCT during the LOT. SCT is operationalized as >1 procedure code for an SCT. A single autologous SCT is one that is not part of a valid tandem pair. Single autologous SCTs and tandem SCTs are not unplanned SCTs and are considered a continuation of the line of therapy.		Autologous HSCT identified from T_MEDICAL procedure codes (PROG_CD, BILL_PROG_CD) matching SCT code list entries for AUTO type. Claims grouped by 14-day windows from T_MEDICAL_FST_DT. Gap validation (60 days) and tandem detection (180 days) applied to grouped dates.		JM 06Apr 2026
	LOT1_HSCT_AUTO_TAND_FL	Flag: base period had tandem valid autologous HSCT		Patients will be classified as having a planned autologous tandem SCT if they receive a second autologous SCT more than or equal 60 days and less than or equal 180 days after the first planned autologous SCT within the same line of therapy (LOT). If the interval between the two autologous SCTs falls outside this window, the sequence will be considered unplanned. An autologous SCT followed by an allogeneic SCT will not be considered planned, because allogeneic SCT is typically follow failed autologous SCTs.		Tandem detection uses T_MEDICAL procedure codes for both AUTO events. The 180-day window is applied to grouped AUTO dates (after 14-day windowing and 60-day gap validation). ALLO events between AUTO_DT_1 and AUTO_DT_2 checked via T_MEDICAL.		JM 15Apr 2026
FU_PD	LOT1_TX_AUTO_MAX_DT	Latest date of patient's valid HSCT		This variable captures the date of the second valid tandem autologous (AUTO_SCT) — where a tandem is two AUTOs >= 60 and <= 180 days apart — or, if only a single AUTO occurred, the date of that single AUTO.		Derived from T_MEDICAL procedure code dates for autologous HSCT, after 14-day windowing and 60-day gap validation.		JM 15Apr 2026

For this LOT algorithm, the team has decided not to define maintenance therapy as its own LOT but rather to flag the inclusion of LOT1 regimens that include potential valid maintenance therapies/therapy combinations.

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PERMISSIBLE_SUBS_EFFECT	Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT; such permissible substitutions are considered within the same line, which is named according to the drug given for the longest duration.	Substitutions tracked via CL_MMA_ROLLUP (Tab 40) which maps medication abbreviations. The permissible_subs reference table pairs original_med with substitute_med. When a new medication claim appears in T_MEDICAL_RX, it is checked against permissible_subs before being classified as a new medication addition.						
LOT1_TX_ENDDATE_REASON (AUTO/ALLO/CART)	SCT end reason type (AUTO/ALLO/CART)	The type of stem cell transplant (SCT) determines how LOT1 ends. In the case of an autologous SCT, an unplanned transplant ends the current line of therapy the day before the SCT date. For an allogeneic SCT, the current LOT1 ends the day before the allogeneic SCT date, and a new LOT1 begins from the date of the transplant. CART: cellular therapy infusions are classified as their own line of therapy, with the preceding LOT ending the day before the CART infusion date; however, if a new oncology agent is introduced within 45 days of the CART infusion, that agent is incorporated into the CART LOT rather than initiating a new line of therapy.						JM 06Apr 2026
LOT1_TX_AUTO_DT_1/ LOT1_TX_AUTO_DT_2	Dates of first and second valid autologous HSCT	Planned Tandem SCT: Patients will be classified as having a planned tandem SCT if they receive at least two autologous SCTs >= 60 to <= 180 days apart. 'SCT is operationalized as: 1 procedure code for an SCT; Tandem SCTs are two SCTs >= 60 to <= 180 days apart. These are considered planned and a continuation of the line of therapy. Codes for SCTs are in the programming specification document.'						JM 06Apr 2026
LOT1_BASE_DISCON_DT	LOT1 base period discontinuation date	Discontinuation of the regimen occurs when all MM agents in the LOT are stopped, either: without initiation of a new MM agent (all agents have reached their run-out dates and no further MM administrations/dispenses are observed), or with initiation of a new MM agent that was not present in the induction regimen. 'The LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available.'						JM 06Apr 2026
LOT1_BASE_1ST_ADD_MED_DT/ LOT1_BASE_1ST_ADD_MED	Date and name of first non-induction medication added	If the current LOT is immediately interrupted by a new agent (the new agent starts on or before the run-out date of the current regimen), the LOT end date is the day before the first administration/dispense date of the new agent. Permissible substitutions (biologic reference product with biosimilars) do not advance the LOT.						JM 06Apr 2026
INDUCTION_WINDOW_DAYS	Induction regimen identification window (days)	The LOT1 regimen includes all MM therapies received within 60 days (Madduri, 2021) on and following the LOT1 start date. For LOT2-LOT5, the LOT regimen includes all MM therapies received within 30 days on and following the LOT start date.						JM 06Apr 2026
ALLO_ALWAYS_ENDS_LOT	Allogeneic SCT always ends current LOT (protocol rule)	Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. An autologous SCT followed by an allogeneic SCT will not be considered planned because an allogeneic SCT typically follows failed autologous SCTs. Rule 2 (SCT-CART events): Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other MM therapies included in that LOT.						JM 06Apr 2026
CART_45D_CONSOLIDATION	CART-45-day supportive agent consolidation rule	CART: cellular therapy infusions are classified as their own line of therapy. Oncology therapies administered within 45 days of the CART-1 infusion are consolidated as part of the CART-1 LOT.						JM 15Apr 2026
NAMING_DISCREPANCY_SCT_VS_HSCT	Variable naming: spec uses HSCT prefixes code uses SCT prefix	The protocol uses 'SCT' (Stem Cell Transplant) terminology consistently throughout. 'Stem cell transplant (SCT)'. The original program spec uses 'HSCT' (Hematopoietic Stem Cell Transplant) prefix for variable names. Both abbreviations refer to the same medical concept.						JM 06Apr 2026
		The study does NOT derive a separate standalone maintenance period or maintenance regimen. This flag records whether LOT1 contains a valid maintenance-approved subset. It does NOT create a maintenance start date, maintenance end date, or LOT-ending event.						
		Set contains_mnt_reg = 1 when BOTH of the following are true for the LOT1 induction regimen: (a) The induction regimen contains at least one valid mono-maintenance agent OR a valid dual-maintenance combination (see lists below). AND (b) At least one additional agent is present outside that qualifying maintenance subset, acting as an anchor.						
		Otherwise contains_mnt_reg = 0.						
		Valid mono-maintenance agents:						
								OK18Apr2026

1. The first step is to identify the problem or question that needs to be answered. This involves understanding the context and the specific requirements of the task.